

QUESTION 1 (20 MARKS)

a) Given the following `ArrayList`, and `Employee` ADTs:

ADT ArrayList**Methods:**

```
ArrayList<T>()
T get(int i)      //get item at position i
void add(T item)  //add new item
int size()        //get size
```

ADT Employee**Fields:**

```
String name
int yearService
String gender
double salary
```

Methods:

```
Employee()
Employee(String n, int y, String g, double s)
int getYearService()
String getGender() //return "Male" or "Female"
double getSalary()
void setSalary(double s)
String toString() //return all information of an employee
```

Assuming that all information about employees are stored in an `ArrayList` object named `empList`, write the code segments using Java to do the following tasks:

- i. Count the number of female and male employees and display whether there is more female or male being employed.

Answer:

(4 marks)

```
Employee emp;
int m =0,f=0;
for(int i =0;i<empList.size();i++){
    emp=empList.get(i);
    if(emp.getGender().equals("Female"))
        f++;
    else
        m++;
}
if(f>m)
    System.out.println("Female is more employed: " +f);
else
    System.out.println("Male is more employed: " +m);
```

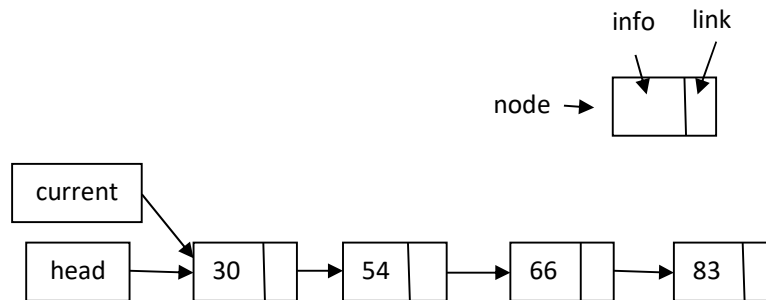
- ii. Update the salary of employees by 10% if they have worked for more than five years and their current salary is below RM5000. Display all the updated employee information. After the update, display the number of employees whose data has been changed.

Answer:

(6 marks)

```
int u =0;
for(int i =0;i<empList.size();i++){
    emp=empList.get(i);
    if(emp.getYearService()>5 &&emp.getSalary<5000){
        emp.setSalary(emp.getSalary()*1.1);
        System.out.println(" Number of updated employee: "+ emp.toString());
        u++;
    }
}
System.out.println(" Number of updated employee: "+ u);
```

b) Given the following linked list, write java code segment that:



- i. Create new node with value 77. Then add the new node at the third position of the linked list.

Answer:

(4 marks)

```
node newNode = new node(77);
current =current.link;
newnode.link = current.link;
current.link = newNode;
```

- ii. Delete node 54 from the linked list.

Answer:

(1 mark)

```
current = head;
current.link= current.link.link;
```

- c) Explain definitions of doubly linked list and circular linked list. Use figures to support your answer.

Answer:

(5 marks)

Doubly linked list is linked list that allows for movement front and back within the list

Circular linked list is linked list where the end of the list is connected to the

first node.

QUESTION 2 (10 MARKS)

a) Given the following code segment on Stack:

```
Stack st=new Stack();
int n=3;
while(n<10){
    st.push(n); //line 1
    n++;
    if(n%2!=0)
        st.pop(); //line 2
}
```

Draw the content of a stack for each push() and pop() operation at //line 1 and //line 1

Answer:

(5 marks)

3

4
3

3

5
3

6
5
3

5
3

7
5
3

8
7
5
3

7
5
3

9
7

5
3

b) Evaluate the following expression:

$$x = a * (b + c / d) - e$$

where a =5 b= 2 c=35 d= 7 e= 8

Answer:

X = abcd/+*e-

(5 marks)

			7					
		35	35	5				
	2	2	2	2	7		8	
5	5	5	5	5	5	35	35	27
5	2	35	7	/	+	*	8	-

QUESTION 3 (10 MARKS)

a) Given the following Queue and House ADTs:

Queue ADT
<pre> public Queue() public void enqueue(House) public House dequeue() public House front() public Boolean isEmpty() public House top() </pre>
House ADT
<pre> private int houseNo private String houseType private double price public House(int,String,double) public int getHouseNo() public String getHouseType()//terrace, semiD, bungalow public double getPrice() </pre>

Write Java program segment to perform the following operations:

- Create a Queue object for Megah Enterprise named `MegahHouse` and input 8 `House` objects into the queue. Ask user for input.

Answer:

(3 marks)

```

Queue MegahHouse = new Queue();
for(int i=0;i<8;i++){
    int hn = sc.nextInt();
    String ht = sc.next();
    double hp = sc.nextDouble();
    MegahHouse.enqueue(new House(hn,ht,hp));
}

```

- Check whether a house is price over RM500000 and its type is terrace. Give 40% discount and update the new price. Store the updated house into new queue object called `discountedHouse`.

Answer:

(5 marks)

```

Queue discountedHouse = new Queue();
while(!MegahHouse.isEmpty()){
    House m = MegahHouse.dequeue();
    if(m.getPrice()>500000 && m.getType().equals('terrace'){
        m.setPrice(m.getPrice*0.6);
        discountedHouse.enqueue(m);
    }
}

```

b) Trace the following code segment:

```

Queue<Integer> q = new LinkedList<>();
q.enqueue(55);
q.enqueue(23);
if(q.top()%5)
    q.dequeue();
q.enqueue(31);

```

Answer:

(2 marks)

55

55 23

23

23 31

QUESTION 4 (10 MARKS)

a) Describes the difference between directed and undirected recursive methods.

Answer:

(2 marks)

Directed the method call is in the method where undirected is the method call another method and the called method call the first method.

b) Given the following codes:

```

public static boolean XYZ(String str, int left, int right) {
    if (left >= right)
        return true;
    if (str.charAt(left) != str.charAt(right))
        return false;
    return XYZ(str, left + 1, right - 1);
}

```

i. Trace the program if for the following method call, XYZ("racecar",0,6).

Answer:

(4 marks)

```

XYZ("racecar",0,6) = XYZ("racecar",1,5)
XYZ("racecar",1,5) = XYZ("racecar",2,4)
XYZ("racecar",2,4) = XYZ("racecar",3,3)
XYZ("racecar",3,3) = true

```

ii. Rewrite the code using iterative method.

Answer:

(4 marks)

```
public static boolean XYZ(String str, int left, int right) {  
    while(left<right){  
        if(str.charAt(left)!=str.charAt(right)  
            return false;  
        left++;right--;  
    }  
    return true;  
}
```

END OF QUESTION PAPER